

# Anaya Koirala

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## Education

Gettysburg College — B.S Computer Science and B.A Philosophy | GPA: 3.870

May 2028

## Professional Experience

**Software Tutor** - Mathematics Department, Gettysburg College

Jan 2026–Present

- Tutor students in tools and softwares for quantitative courses across campus.
- Aid with debugging and problem solving in code based coursework.
- Specialized in Python, LaTeX, Java, WolframLang and statistical softwares.

**Teaching Assistant for CS I and II** — Computer Science Department, Gettysburg College

Jan–Dec 2025; Aug 2026–Present

- Supported approx. 30 students on core programming concepts, OOP, GUI, Data Structures
- Conducted weekly office hours for debugging, coursework, and technical issues

**Software Developer** — Kha Esports, Kathmandu, Nepal

Jun 2025–Aug 2025

- Developed website for hosting, managing, and participating in e-sports tournaments
- Worked on a multiplayer ludo game with for 64 simultaneous players
- Tools Used: FastAPI, MongoDB, React.js, Express.js, nginx

## Relevant Coursework

- **Mathematics of Computation:** Combinatorics, Proofs, Graph Theory, Logic.
- **Calculus III:** Multivariable and Vector Calculus with Applications.
- **Linear Algebra:** Applications in compression, data analytics, Markov chains, IFS.
- **Artificial Intelligence:** Machine Learning, Reinforcement Learning, Game-Tree Search, Bayesian Reasoning, Markov Chains.

## Technologies

**Languages:** Python, Java, Rust, C#, OCaml, F#, Shell, SQL, Haskell

**Tools:** Linux, BSDs, Windows, LaTeX, JUnit, NumPy, Sage, JavaFX

## Projects

**SubHunt** — Audio-based PvE video game for blind players built in Godot

[GitHub](#)

- Worked in game design and development for the [Games for Blind Gamers 4](#) game jam.

**MelodyTron** — CS 371: Intro to AI Final Project

- Generate folk songs from an initial seed melody using latest machine learning tools.

[Website](#)

**Musical-Style-Replicating-LSTM** — IBDP Extended Essay project

[GitHub](#)

- Self-directed highschool AI research project in emulating a composer's musical style

**ireu** - A classifier and database of REU programs that accept international students

[Website](#)

- Built in F# and Python using Web Scraping and LLM-based tools.

**calscan** - Mobile app to directly convert images of posters and flyers into a calendar event.

[Website](#)

- Built in Dart with Flutter, ML Kit Text Recognition and iCalendar RFC 5545.

**codeDP** — Desktop code editor for IB Computer Science pseudocode

[GitHub](#)

- Built using Python and QT6

## Academic Work

- “Is There a Collective in *Zhuangzi*’s Freedom ?” — MCCT Pre-Conference Student Panel 2026, Gettysburg College

Apr 2026

## Awards

- [Andrew Marsh Memorial Award](#) — Gettysburg College

Oct 2025

## Extra and Co-Curricular Activities

- **ACM Club** — Webmaster Team and Outreach Chair
- **The Gettysburg Cinematics** — Senate Representative
- **Quiz Bowl Club** — Senate Representative
- **Amnesty International Club** — Treasurer
- [The Gettysburgian](#) — Writer

Oct 2025 – Present

Sept 2025 – Present

Apr 2024 – Present

Sept 2025 – Present

Jan 2025– Present